

Neuromorphic Hebbian Learning with Magnetic Tunnel Junction Synapses

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Abstract—The use of analog resistance states for storing weights in neuromorphic systems is impeded by stochastic writing, weights drifting over time through stochastic processes, and limited endurance that reduces the precision of synapse weights. In [1], we propose and experimentally demonstrate neuromorphic networks that provide high-accuracy inference thanks to the binary resistance states of magnetic tunnel junctions (MTJs), while leveraging the analog nature of their stochastic spin-transfer torque (STT) switching for unsupervised Hebbian learning. We performed an experimental demonstration of a neuromorphic network directly implemented with MTJ synapses, for both inference and spike-timing-dependent plasticity learning. We also demonstrated through simulation that the proposed system for unsupervised Hebbian learning with stochastic STT-MTJ synapses can achieve competitive accuracies for MNIST handwritten digit recognition. By appropriately applying neuromorphic principles through hardware-aware design, the proposed STT-MTJ neuromorphic learning networks provide a pathway toward artificial intelligence hardware that learns autonomously with extreme efficiency.

Index Terms—Hebbian learning, magnetic tunnel junction, neuromorphic computing, spin-transfer torque, stochastic switching.

I. INTRODUCTION

Energy-efficient artificial intelligence hardware is increasingly turning to spiking neural networks (SNNs) [2], which carry information through pulses and store synaptic strengths in non-volatile resistive devices [3]. Memristors [4] and phase-change cells have been investigated as analog synapses, but each suffers from imprecise programming, drift of stored values, and finite write endurance. These limitations have motivated a shift toward devices restricted to two stable levels [1]. MTJs naturally provide such binary states through their parallel (P) and anti-parallel (AP) configurations [5], and their inherently stochastic STT writing supplies an analog-like resource appropriate for plasticity rules [6]. Earlier theoretical proposals considered MTJs as stochastic synapses [7], and prior experiments by Goodwill [8] and by Jung [9] explored MTJ-based neuromorphic hardware at scale, but those efforts included neither unsupervised learning nor a direct implementation of vector-matrix multiplication with single-MTJ synapses [1]. The work in [1] addresses that gap by combining inference and unsupervised learning in one hardware demonstrator and projecting performance through detailed device-level simulations.

II. BINARY MTJ INFERENCE NETWORK

A. Array and Bipolar Weight Encoding

The reported study used eight in-plane STT-MTJs arranged in a 4×2 crossbar, with sixteen probes contacting the top and bottom electrodes of the eight junctions. Each device stores one bit through its tunneling magnetoresistance: the P state provides high conductance and the AP state provides low conductance. Inputs are applied as binary column voltages, and row currents act as dendritic sums [1]. To emulate signed weights despite the

modest tunneling magnetoresistance, a constant proportional to the count of active inputs is subtracted from each row, which needs only minimal digital logic per column [1].

B. Pattern Matching

Programmed with two reference 2×2 images, the array was probed with all sixteen possible inputs. Post-subtraction currents reproduced the expected Hamming distances and sorted cleanly into five well-separated bins. Residual spread was attributed to power-supply variation and differences in probe connectivity rather than intrinsic device spread, and substantially smaller variation is anticipated once future arrays are directly co-integrated with peripheral CMOS [1], [10].

III. UNSUPERVISED HEBBIAN LEARNING

A. Stochastic STDP Rule

The binary synapses are switched probabilistically, with distinct switching probabilities for potentiation and depression pulses. Pulse magnitude and duration thus program a graded probability rather than a graded resistance, yielding system-level analog plasticity from strictly two-state devices [1].

B. Neurons and Homeostasis

Surrounding CMOS implements leaky integrate-and-fire neurons followed by a winner-take-all stage [2]. Firing thresholds are adjusted dynamically: a neuron that fired within a recent window has its threshold raised, while a neuron quiet for a longer window has its threshold lowered. Following a winning event, potentiation pulses target synapses whose pre-synaptic input was active in the recent past, and depression pulses target the remaining synapses on that row [1]. The combination of stochastic plasticity with adaptive thresholding enforces a bal-

anced specialization across output neurons without external supervision.

IV. EXPERIMENTAL CLUSTERING

The 4×2 array was reconfigured with a manual switch board that emulated the surrounding CMOS by routing the fixed layers of each junction to inference, potentiation, or depression voltages and the dendrites to the source-measure units or ground as appropriate [1]. Two distinct 2×2 patterns were then presented. Pulse parameters set switching probabilities to roughly 35% (AP $\rightarrow$ P) and 30% (P $\rightarrow$ AP). Starting from identical row states, the network was repeatedly stimulated with the patterns in random order, each presentation ending at a firing event. Within nine presentations, the two output neurons specialized onto the two distinct patterns, confirming that random STT events combined with local plasticity drive cluster formation [1]. The experiment thus validates a key prediction of stochastic-synapse theory in physical hardware rather than only in simulation.

V. MNIST SCALING AND ENERGY EFFICIENCY

Behavioral simulations on a binarized MNIST set used 784 input neurons and treated output-layer size and synaptic redundancy r (one to eight junctions per pixel) as design parameters. Following unsupervised training, output neurons were assigned digit labels from their firing statistics on the training set, and accuracy was evaluated on the standard test set. With 10,000 output neurons and $r = 8$, accuracy reaches roughly 90%, on par with single-layer SNN results for multilevel memristive synapses [4] but without idealized analog programming. A complementary analog-weight simulation with unrealistic perfect analog weights indicates the realistic binary network needs about twice the output count to match the unrealistic analog case ($\lambda \approx 1.96$) [1]. This modest size penalty is offset by the feasibility, the elimination of write-precision errors, and the long-term stability of the stored magnetic states [5], [10]. For a 256×256 layer, inference is estimated at about 633 TOPS/W—roughly $6 \times$ better than memristor macros and over $140 \times$ better than the NVIDIA H100 Tensor Core GPU at 7 nm [1]—while a stochastic update with $V_{\text{write}} = 0.8$ V and $\tau_{\text{write}} = 5$ ns consumes about 20.5 pJ per Hebbian attempt, versus several nJ for an equivalent CMOS-based weight refresh [1]. Parasitic-line and peripheral overheads, which dominate large analog crossbars, are reduced by the high on-state conductance and binary operation of the junctions.

VI. DISCUSSION AND OUTLOOK

Several broader implications follow from these results. First, the use of a commercially manufactured STT magnetic memory process [10] suggests a relatively short path from laboratory demonstration to integration alongside standard logic. Second, by reframing device randomness as a useful resource rather than a defect, the architecture more efficiently leverages the natural characteristics intrinsic to magnetic devices. Third, because the plasticity rule depends only on locally available signals, the

approach is well suited to edge-deployed systems where data movement to a host is prohibitively expensive.

VII. CONCLUSIONS

The work in [1] shows that binary STT-MTJ devices can simultaneously support accurate dot-product inference and probabilistic Hebbian learning, removing the long-standing tension between weight precision and analog-style plasticity. Hardware experiments verify both functions on a small array, and simulations indicate that the architecture extrapolates to application-relevant sizes with attractive efficiency. Compatibility with commercially manufactured STT magnetic memory [10] suggests a short path toward integrated on-chip learning systems for autonomous edge devices, and positions magnetic synapses as a strong candidate substrate for future unsupervised neuromorphic processors.

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